

activity 4.2

biopsychology bingo

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CONCEPT

Students will participate actively in a review game of key terms and concepts in biopsychology.

MATERIALS

You will need Biopsychology Bingo questions, Biopsychology Bingo terms list, bingo cards, an overhead projector or PowerPoint slide, and candy prizes (optional). The following websites provide free sample customized or blank bingo cards:

www.print-bingo.com

www.teach-nology.com/web_tools/materials/bingo (requires membership)

INSTRUCTIONS

Provide students with blank or customized bingo cards and present the biopsychology bingo terms either on an overhead, the white board, or a PowerPoint slide. If you are using a blank card, ask students to choose any 24 terms from the provided list to fill in their own bingo card. You may also choose to create your own varied bingo cards to present to the students. For each round, the teacher reads questions from the provided biopsychology bingo list, and students attempt to complete their bingo cards. After a student shouts bingo, have them read their answers and award them candy if they are a winner (always check with students about possible allergies before distributing food in class). Before moving on to another round, be certain to review the correct answers.

BIOPSYCH BINGO QUESTIONS

What is/are the...

1. Connection between cerebral hemispheres (corpus callosum)
2. Part of the limbic system, important in the fear response (amygdala)

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3. Collection of forebrain structures mainly involved in emotion (limbic system)
4. Structures most involved in producing speech, located in the left hemisphere for most people (Broca's area)
5. Brain and spinal cord (central nervous system)
6. Sensory and motor neurons found outside the central nervous system (peripheral nervous system)
7. Specialized cells that communicate information in the nervous system (neurons)
8. A threadlike structure that action potential travels along (axon)
9. Postsynaptic structures in a neuron that receive messages (dendrites)
10. Small gap between neurons crossed by neurotransmitters (synapse)
11. Coating that may develop on an axon that speeds neural transmission (myelin)
12. Neurotransmitter whose imbalance is involved in schizophrenia and Parkinson's disease (dopamine)
13. Neurotransmitter involved in mood regulation, target of many antidepressant drugs (serotonin)
14. Major inhibitory neuroregulator (GABA)
15. Major excitatory neuroregulator (glutamate)
16. Neurotransmitter most critical for memory, also used for muscle contraction (acetylcholine)
17. Cerebral region that is the primary processing area for vision (occipital lobe)
18. Cerebral region that is the primary processing area for hearing (temporal lobe)
19. Cerebral region that is the primary processing area for higher brain functions and motor messages (frontal lobe)
20. Cerebral region that is the primary processing area for skin senses (parietal lobe)
21. Division of the peripheral nervous system that includes sympathetic and parasympathetic branches (autonomic)
22. Connection between the brain and the peripheral nervous system (spinal cord)
23. Representation on the brain of the relative sensitivity of various regions of sense receptors (sensory homunculus)
24. The body's natural opiate-like pain killer (endorphin)
25. Area of the limbic system that is important in storage of memories (hippocampus)
26. Brain structure that regulates many important functions through stimulation of the pituitary "master gland" (hypothalamus)
27. Structure that relays information from all senses except smell (thalamus)
28. The electrical potential at which an action potential fires (threshold)
29. Hormones that are used by the nervous system to communicate messages through the body (endocrine system)
30. Structure at the back of the brain involved in balance, fine motor skills, and sequencing (cerebellum)
31. Brain-imaging technique that uses powerful magnets to create sharp pictures (MRI)
32. Measurement that tracks brain waves (EEG)
33. Interneuron loop that results in automatic action before brain processing (reflex)
34. Nervous system component that provides support for neurons (glia)
35. Thin layer of cells covering the forebrain, site of most brain activity (cortex)
36. Division of brain structures that include functions most critical to living (hindbrain)

BIOPSYCH BINGO TERMS

Review the following terms dealing with the nervous system.

Choose any 24 from the list to fill in your Biopsych Bingo card in any order.

1. Acetylcholine
2. Amygdala
3. Axon
4. Autonomic nervous system
5. Broca's area
6. Central nervous system
7. Cerebellum
8. Corpus callosum
9. Cortex
10. Dendrite
11. Dopamine
12. EEG
13. Endocrine system
14. Endorphin
15. Frontal lobe
16. GABA
17. Glia
18. Glutamate
19. Hindbrain
20. Hippocampus
21. Hypothalamus
22. Limbic system
23. MRI
24. Myelin
25. Neuron
26. Occipital lobe
27. Parietal lobe
28. Peripheral nervous system
29. Reflex
30. Sensory homunculus
31. Serotonin
32. Spinal cord
33. Synapse
34. Temporal lobe
35. Thalamus
36. Threshold